

Science

1. (4) +0.5
2. (2) outer surface of the cornea
3. (4) Virtual, erect and enlarged
4. (1) Both A and R are ~~true~~ true and R is the correct explanation of A.
5. (3) Assertion is true but reason is false
6. The two most common defect of vision are hypermetropia and myopia. Myopia can be corrected by using a concave lens and hypermetropia can be corrected by using convex lens.
7. (a) Water is optically denser medium compared to air. The light from the part of pencil which is immersed in water travels from water to air causing bending of light

(b) Given $\Rightarrow$ magnification = +2.0
 $u = ?$
 $v = 20 \text{ cm}$

Using $m = \frac{v}{u}$

$$2.0 = \frac{20}{u}$$

$$u = \frac{20}{2.0} = 10 \text{ cm}$$

Object distance from the lens is 10cm

8. Given $\Rightarrow f = +15\text{cm}$
 $\Rightarrow u = -10\text{cm}$

using $\Rightarrow \frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\Rightarrow \frac{1}{+15} = \frac{1}{v} + \frac{1}{(-10)}$$

$$\frac{1}{v} = \frac{1}{15} + \frac{1}{10}$$

$$\frac{1}{v} = \frac{2+3}{30} \Rightarrow \frac{5}{30}$$

$$v = \frac{30}{5} = +6\text{cm}$$

The image is formed behind the mirror, It is virtual and erect image

9. Given $\Rightarrow u = -10$
 $m = -3$, $v = ?$, $f = ?$

using, $m = \frac{v}{u}$

$$-3 = \frac{v}{-10}$$

$$+30 = v$$

Image distance from the lens is 30 cm.

using $\Rightarrow \frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

$$\frac{1}{f} = \frac{1}{30} - \frac{1}{(-10)}$$

$$\frac{1}{f} \Rightarrow \frac{1+3}{30} = \frac{4}{30}$$

$$f = 7.5 \text{ cm}$$

The focal length of the lens is 7.5 cm

10. Myopia can be corrected by using concave lens. The lens must be placed that an object at infinity forms its image of on far point.

$$\text{Hence, } f = -80 \text{ cm}$$

$$f = -0.8 \text{ m}$$

$$P = \frac{1}{f}$$

$$P = \frac{1}{-0.8 \text{ m}} \Rightarrow -1.25 \text{ D}$$

11. (a) Convex lens

- (b) The image is real and inverted in nature hence $\frac{v}{u}$ (magnification) is negative.

(c) Using, $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$

$$\frac{1}{20} = \frac{1}{v} - \frac{1}{-24}$$

$$\frac{1}{v} = \frac{1}{20} - \frac{1}{24}$$

$$\frac{1}{v} = \frac{24 - 20}{480}$$

$$\frac{1}{v} = \frac{4}{480}$$

$$v = 120 \text{ cm}$$

Screen should be placed 120 cm away from the slide projector

12 (a) Refractive index is defined as the ratio of the speed of light in vacuum to its speed in a specific medium.

$$\text{refractive index}(n) = \frac{\text{speed of light in vacuum} \rightarrow c}{\text{speed of light in medium} \rightarrow v}$$

(b) Given $\rightarrow$ refractive index of water $\rightarrow n_w = \frac{4}{3}$
refractive index of glass $\rightarrow n_g = \frac{3}{2}$

speed of light in air $= 3 \times 10^8 \text{ m/s}$
speed of light in glass $= 2 \times 10^8 \text{ m/s}$

$$n_g = \frac{c}{v_g}$$

$$c = v_g \times n_g$$

$$c = \frac{2 \times 10^8 \times 3}{2}$$

$$c = 3 \times 10^8 \text{ ms}^{-1}$$

speed of light in air $\Rightarrow 3 \times 10^8 \text{ m/s}$

$$n_w = \frac{c}{v_w}$$

$$v_w = \frac{c}{n_w}$$

$$v_w = \frac{3 \times 10^8 \times 3}{4}$$

in water
speed of light $\Rightarrow 2.25 \times 10^8 \text{ m s}^{-1}$